



BRAINWARE UNIVERSITY

School of Engineering

Department of Computer Science & Engineering – Cyber Security & Data Science

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

PBL ASSIGNMENT – 4

Multi-department University Network Design Using RIP and OSPF

Course Code: BTD50112

Course Name: Computer Networks

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Title: Multi-department University Network Design Using RIP and OSPF.

Objective

Design a multi-router university network and evaluate RIP and OSPF based on routing table formation, convergence, hop count, path selection, and Network scalability.

Real Life Connection

A university has three geographically separately building like: academic, administrative, and research. Each building has its own LAN. Routers are used to connect the buildings. The network administrator wants to compare RIP and OSPF for dynamic routing.

Target

1. Design a 3-router network in Cisco Packet Tracer
2. Configure IP addresses.
3. Configure RIP and OSPF
4. Examine routing tables & also observe how routing changes
5. Determine the preferred protocol for the university network.

Group Members

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Objective

The objective of this project-based learning (PBL) assignment is to design and simulate a multi-department university network consisting of three geographically separate LANs — Academic, Administrative, and Research — interconnected through routers, and to critically evaluate two dynamic routing protocols, RIP (Routing Information Protocol) and OSPF (Open Shortest Path First), on the same underlying campus network.

Specifically, this project aims to:

- Design a scalable 3-router, 3-LAN university network topology in Cisco Packet Tracer that mirrors a real campus environment.
- Perform structured IP addressing and subnetting for each department LAN as well as for the inter-router backbone links.
- Configure RIP (version 2) on one topology and OSPF (single area, Area 0) on a second topology, and verify end-to-end reachability using ping and traceroute.
- Examine and compare the routing tables generated by each protocol, including the source of each route, administrative distance, and metric.
- Observe protocol behaviour with respect to convergence time, hop-count limitations, and path-selection logic when multiple paths exist.
- Analyse the scalability, efficiency, and suitability of each protocol for a growing university network, and recommend the protocol best suited for long-term deployment.

Solution (Subnetting) & Design (Packet Tracer)

5.1 Addressing Plan & Subnetting

Each department (Academic, Administrative, Research) is assigned an independent Class C LAN so that host addressing within a building can be managed and scaled without affecting other departments. A separate set of subnets is used to interconnect the routers so that LAN broadcast traffic never crosses the backbone. The complete addressing plan is summarised below.

Table 1: Department LAN Subnets

Department / Network	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Network 1 - Academic Bldg	192.168.10.0	255.255.255.0 /24	.1 - .254 (254 hosts)	192.168.10.1
Network 2 - Admin Bldg	192.168.20.0	255.255.255.0 /24	.1 - .254 (254 hosts)	192.168.20.1
Network 3 - Research Bldg	192.168.30.0	255.255.255.0 /24	.1 - .254 (254 hosts)	192.168.30.1

Table 2: Backbone (Inter-router) Subnets - OSPF Topology

Backbone Link	Network Address	Subnet Mask	Router A IP	Router B IP
Network 4 (Router0 - Router1)	10.10.10.0	255.255.255.0 /24	10.10.10.1 (R0)	10.10.10.2 (R1)
Network 5 (Router1 - Router2)	20.20.20.0	255.255.255.0 /24	20.20.20.1 (R1)	20.20.20.2 (R2)

Table 3: Backbone (Inter-router) Subnets - RIP Topology

Backbone Link	Network Address	Subnet Mask	Router A IP	Router B IP
Network 4 (Router0 - Router1)	10.0.0.0	255.255.255.0 /24	10.10.10.1 (R0)	10.10.10.2 (R1)
Network 5 (Router1 - Router2)	20.0.0.0	255.255.255.0 /24	20.20.20.1 (R1)	20.20.20.2 (R2)
Network 6 (Router0 - Router2)	30.0.0.0	255.255.255.0 /24	30.30.30.1 (R0)	30.30.30.2 (R2)

Note: The RIP topology adds a third backbone link (Network 6, Router0 - Router2) so that a triangular, meshed backbone is formed. This deliberately creates two possible paths between the Academic and Research LANs, which is required to observe RIP's hop-count based path selection in Section 6 (Observation).

Table 4: End-host (PC) Addressing

PC / Host	Assigned IP	Subnet Mask	Default Gateway
PC - Academic (x4)	192.168.10.2 - .5	255.255.255.0	192.168.10.1
PC - Admin (x2)	192.168.20.2 - .3	255.255.255.0	192.168.20.1
PC - Research (x4)	192.168.30.2 - .5	255.255.255.0	192.168.30.1

5.2 Network Design in Cisco Packet Tracer

Two parallel topologies were built in Packet Tracer using identical LAN structures (three 2960-24TT switches, each with 2-4 PCs) so that the only variable between the two experiments is the routing protocol and, consequently, the backbone shape needed to test that protocol meaningfully.

5.2.1 OSPF Topology - Linear Backbone

Router0, Router1 and Router2 are connected in a single-area (Area 0) linear chain: Router0 - Router1 - Router2. Each router also connects to its own department LAN through a 2960-24TT switch.

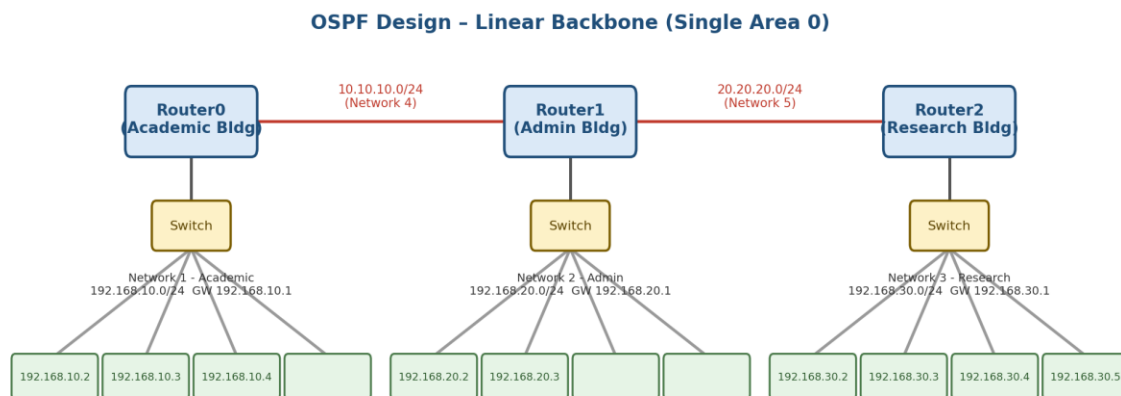


Figure 1: OSPF network design - linear backbone connecting the three department LANs.

5.2.2 RIP Topology - Triangular (Meshed) Backbone

For the RIP experiment, an additional backbone link (Network 6) is added directly between Router0 and Router2, forming a triangle. This provides two independent paths between the Academic and Research networks (a direct 1-hop path and a 2-hop path via Router1), which allows RIP's distance-vector, hop-count based best-path selection to be clearly observed.

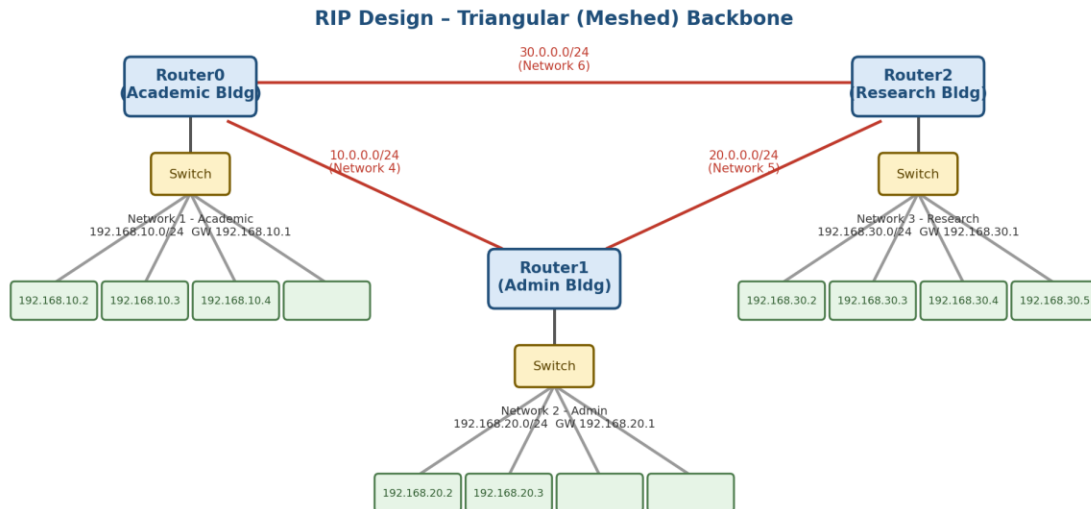


Figure 2: RIP network design - triangular, meshed backbone with redundant paths.

5.3 Router Configuration Summary

The key configuration commands applied on each router are summarised below (interface addressing is applied first in both cases, followed by the routing protocol):

OSPF Configuration (per router)

- interface <backbone-int> -> ip address <network-ip> 255.255.255.0 -> no shutdown
- interface <lan-int> -> ip address <gateway-ip> 255.255.255.0 -> no shutdown
- router ospf 1
- network <lan-network> 0.0.0.255 area 0
- network <backbone-network> 0.0.0.255 area 0

RIP Configuration (per router)

- interface <backbone-int> -> ip address <network-ip> 255.255.255.0 -> no shutdown
- interface <lan-int> -> ip address <gateway-ip> 255.255.255.0 -> no shutdown
- router rip
- version 2
- network <lan-network>
- network <backbone-network>
- no auto-summary

5.4 Packet Tracer Implementation Screenshots

The screenshots below show the final Cisco Packet Tracer implementation of both experiments: the completed RIP and OSPF topologies with their interface IP addresses, followed by the router CLI used to enable each routing protocol on Router0.

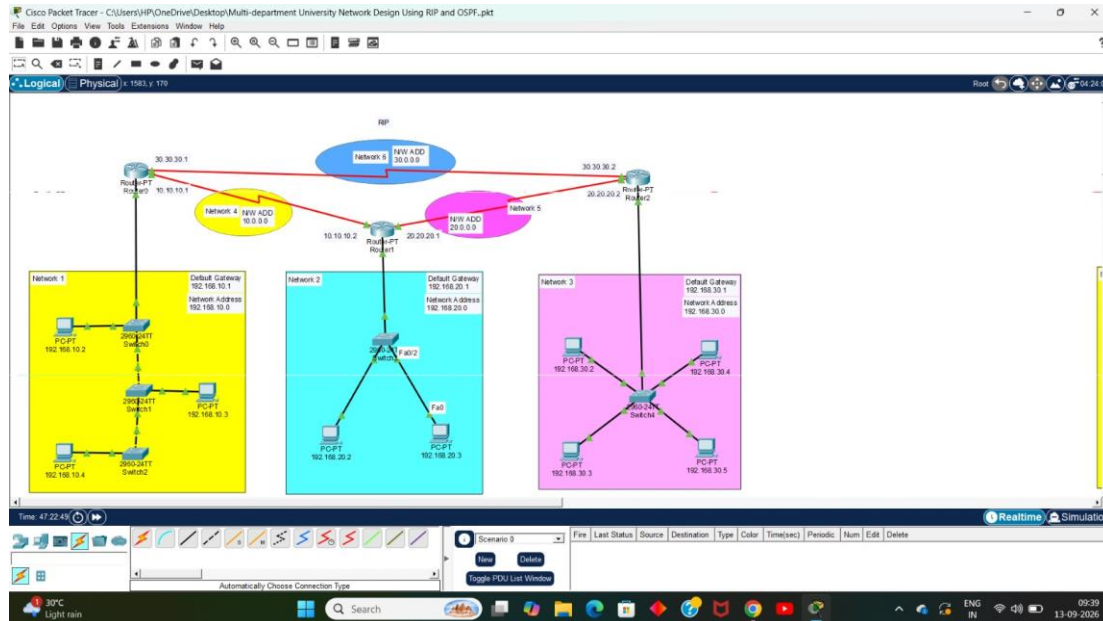


Figure 3: Cisco Packet Tracer - RIP topology with triangular backbone (Networks 4, 5 and 6) and department LANs.

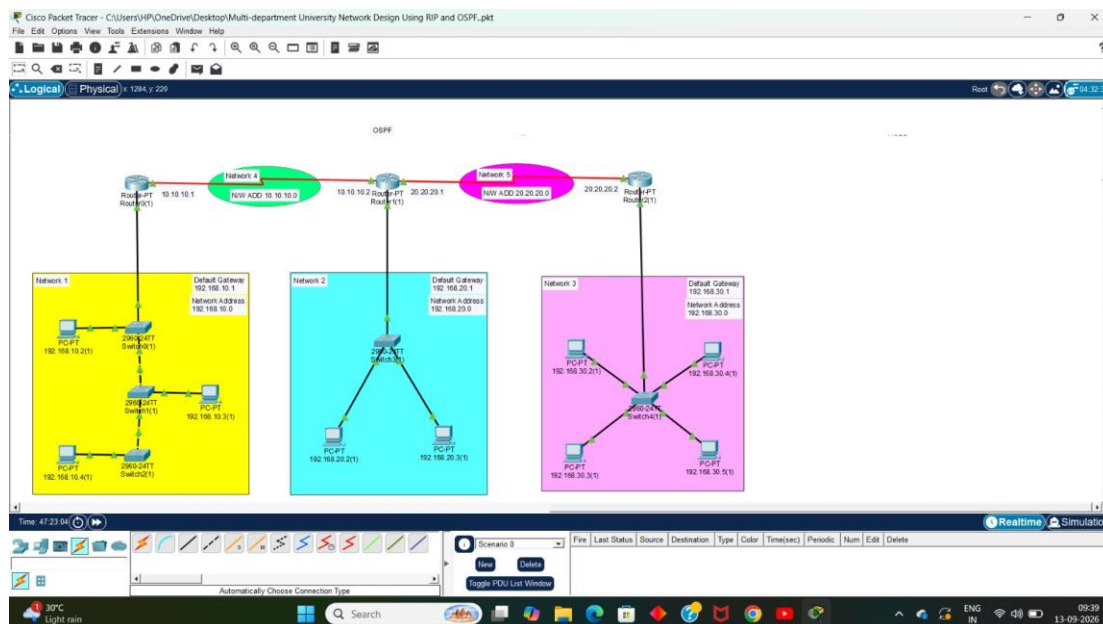


Figure 4: Cisco Packet Tracer - OSPF topology with linear backbone (Networks 4 and 5) and department LANs.

In the OSPF experiment, Router0 was configured with the OSPF process and all five networks (the three department LANs and the two backbone networks) were advertised in Area 0. The console output confirms the OSPF neighbour adjacency moving from LOADING to FULL.

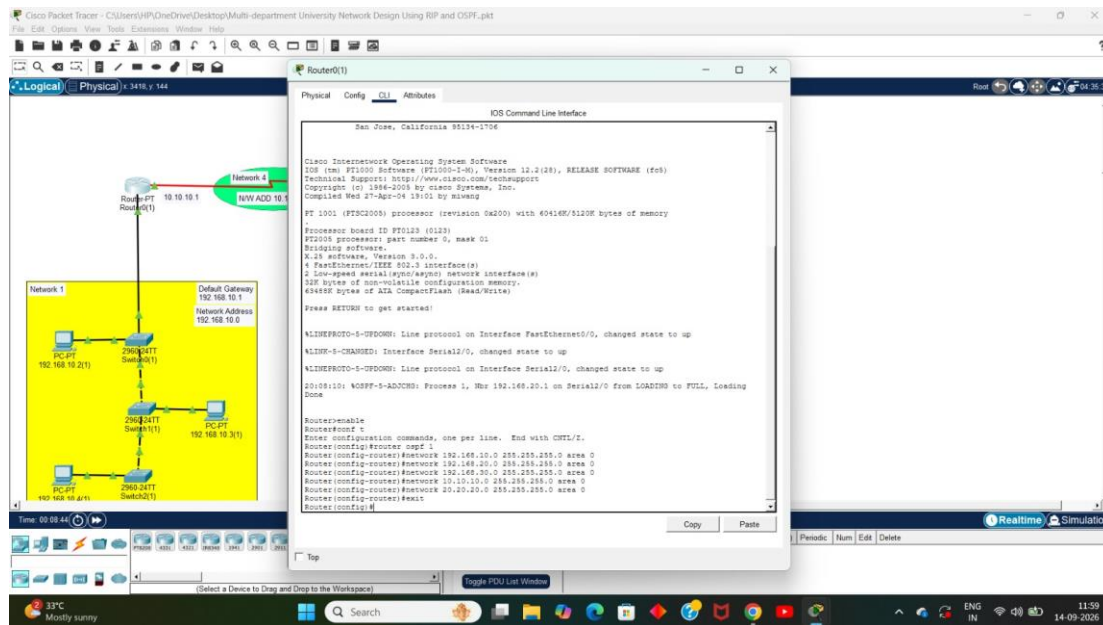


Figure 5: OSPF configuration on Router0 (router ospf 1, all networks in area 0) and adjacency reaching FULL state.

In the RIP experiment, Router0 was configured with the RIP process and its directly connected networks were advertised: the Academic LAN (192.168.10.0) and the backbone networks 10.10.10.0 and 30.30.30.0.

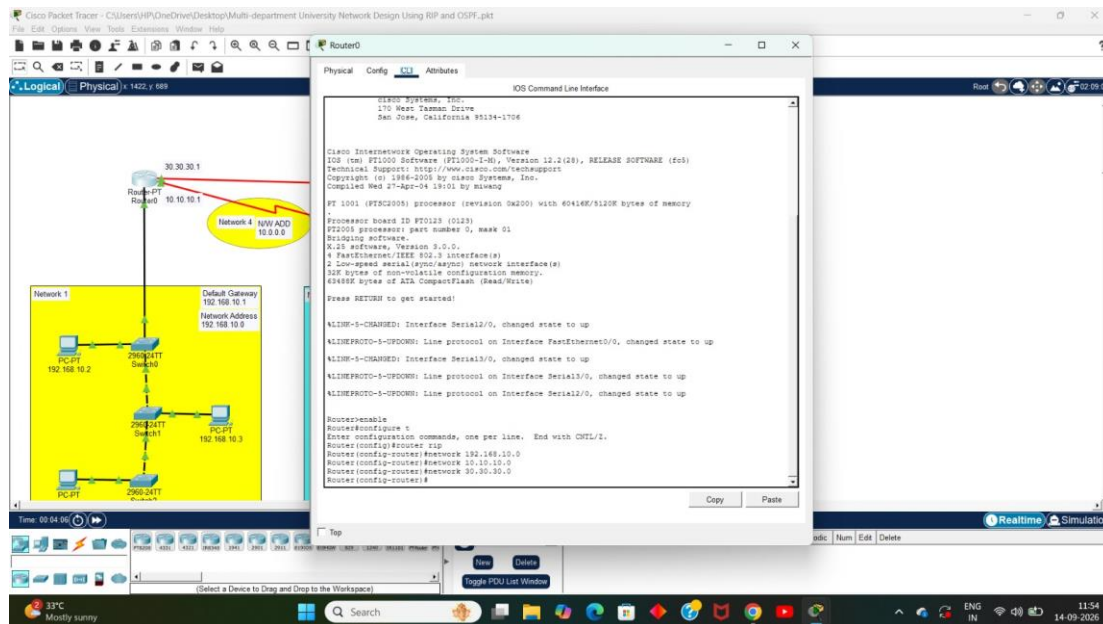


Figure 6: RIP configuration on Router0 - directly connected networks advertised under router rip.

5.5 CLI Configuration Commands

The complete command-line configuration applied on each router in Cisco Packet Tracer is listed below. In every router, the department LAN is connected to FastEthernet0/0 (default gateway of that LAN) and the backbone links use the serial interfaces. The commands are entered from the CLI tab of each router after pressing RETURN to get started.

5.5.1 OSPF Configuration (Linear Backbone)

OSPF process 1 is used with all interfaces placed in Area 0. Wildcard mask 0.0.0.255 is used because every network is a /24.

Router0 - OSPF

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0
Router0(config)#interface FastEthernet0/0
Router0(config-if)#ip address 192.168.10.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface Serial2/0
Router0(config-if)#ip address 10.10.10.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#router ospf 1
Router0(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router0(config-router)#network 10.10.10.0 0.0.0.255 area 0
Router0(config-router)#exit
Router0(config)#end
Router0#copy running-config startup-config
```

Router1 - OSPF

```
Router>enable
Router#configure terminal
Router(config)#hostname Router1
Router1(config)#interface FastEthernet0/0
Router1(config-if)#ip address 192.168.20.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface Serial2/0
Router1(config-if)#ip address 10.10.10.2 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface Serial3/0
Router1(config-if)#ip address 20.20.20.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#router ospf 1
Router1(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router1(config-router)#network 10.10.10.0 0.0.0.255 area 0
Router1(config-router)#network 20.20.20.0 0.0.0.255 area 0
Router1(config-router)#exit
Router1(config)#end
Router1#copy running-config startup-config
```

Router2 - OSPF

```
Router>enable
Router#configure terminal
Router(config)#hostname Router2
Router2(config)#interface FastEthernet0/0
Router2(config-if)#ip address 192.168.30.1 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#interface Serial2/0
Router2(config-if)#ip address 20.20.20.2 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#router ospf 1
Router2(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router2(config-router)#network 20.20.20.0 0.0.0.255 area 0
Router2(config-router)#exit
Router2(config)#end
Router2#copy running-config startup-config
```

5.5.2 RIP Configuration (Triangular Backbone)

RIP version 2 is enabled with auto-summary disabled so that the subnets are advertised exactly as configured. The extra Router0 - Router2 link (Network 6) uses the second serial interface on Router0 and Router2.

Router0 - RIP

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0
Router0(config)#interface FastEthernet0/0
Router0(config-if)#ip address 192.168.10.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface Serial2/0
Router0(config-if)#ip address 10.10.10.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface Serial3/0
Router0(config-if)#ip address 30.30.30.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#router rip
Router0(config-router)#version 2
Router0(config-router)#network 192.168.10.0
Router0(config-router)#network 10.10.10.0
Router0(config-router)#network 30.30.30.0
Router0(config-router)#no auto-summary
Router0(config-router)#exit
Router0(config)#end
Router0#copy running-config startup-config
```

Router1 - RIP

```
Router>enable
Router#configure terminal
Router(config)#hostname Router1
Router1(config)#interface FastEthernet0/0
Router1(config-if)#ip address 192.168.20.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface Serial2/0
Router1(config-if)#ip address 10.10.10.2 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface Serial3/0
Router1(config-if)#ip address 20.20.20.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#router rip
Router1(config-router)#version 2
Router1(config-router)#network 192.168.20.0
Router1(config-router)#network 10.10.10.0
Router1(config-router)#network 20.20.20.0
Router1(config-router)#no auto-summary
Router1(config-router)#exit
Router1(config)#end
Router1#copy running-config startup-config
```

Router2 - RIP

```
Router>enable
Router#configure terminal
Router(config)#hostname Router2
Router2(config)#interface FastEthernet0/0
Router2(config-if)#ip address 192.168.30.1 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#interface Serial2/0
Router2(config-if)#ip address 20.20.20.2 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#interface Serial3/0
Router2(config-if)#ip address 30.30.30.2 255.255.255.0
Router2(config-if)#no shutdown
Router2(config-if)#exit
Router2(config)#router rip
Router2(config-router)#version 2
Router2(config-router)#network 192.168.30.0
Router2(config-router)#network 20.20.20.0
Router2(config-router)#network 30.30.30.0
Router2(config-router)#no auto-summary
Router2(config-router)#exit
Router2(config)#end
Router2#copy running-config startup-config
```

Observation

After configuring both topologies in Cisco Packet Tracer and allowing the protocols to converge, the routing tables of Router0, Router1 and Router2 were examined using the "show ip route" command, and reachability was verified with ping and traceroute between department LANs. The following observations were recorded.

Parameter	OSPF (Linear Backbone)	RIP (Triangular Backbone)
Route code seen in routing table	O (OSPF, AD 110)	R (RIP, AD 120)
Metric used	Cost (based on bandwidth)	Hop count
Routing table formation	Learns Router LSAs, builds full topology (link-state database), then runs SPF (Dijkstra) to compute best paths	Learns entire routing table from directly connected neighbours and periodically exchanges full table (distance-vector)
Convergence speed	Fast - triggered updates as soon as a link-state change is flooded	Slow - relies on periodic updates (every 30 s) and hold-down / timeout timers
Hop-count limit	No hard hop limit (suitable for larger campus networks)	Maximum 15 hops; 16 = unreachable, limiting scalability
Path selection (with redundant link)	N/A in this linear design - only one physical path exists	With the added Router0-Router2 link, RIP selects the 1-hop direct path over the 2-hop path via Router1
Bandwidth utilisation	Uses multicast (224.0.0.5/6) only on link-state changes	Broadcasts / multicasts (224.0.0.9) the full table every 30 seconds regardless of change
Scalability for university growth	High - supports areas, VLSM, and hundreds of routers	Low - flat network only, becomes inefficient beyond ~15 hops

6.1 Routing Table Formation

In the OSPF network, each router first exchanged Hello packets to form neighbour adjacencies over the point-to-point backbone links, then flooded Link-State Advertisements (LSAs) to build an identical link-state database on every router. Each router independently ran the Shortest Path First (SPF) algorithm on this database to compute its routing table, with routes marked with the code 'O' and an administrative distance of 110.

In the RIP network, each router learned about remote networks purely from its directly connected neighbours by periodically broadcasting its entire routing table. Routes appeared with the code 'R' and an administrative distance of 120, with the metric expressed simply as the number of router hops to the destination.

6.2 Convergence

When a backbone link was deliberately shut down during simulation, OSPF re-converged almost immediately because the affected router flooded a new LSA describing the topology change, which every

other router used to instantly re-run SPF. RIP took noticeably longer to converge because it depends on periodic (30-second) update cycles and hold-down timers before removing or replacing a failed route, resulting in a temporary period of stale/incorrect routing information.

6.3 Hop Count & Path Selection

Because the RIP topology includes a third backbone link directly between Router0 and Router2 (in addition to the path via Router1), it was possible to observe RIP's path-selection behaviour: RIP always preferred the single-hop direct path over the two-hop path through Router1, purely on the basis of hop count, even though both paths were realistically equivalent in bandwidth in the simulated network. OSPF's linear backbone had no redundant path in this design, so its cost-based path selection could only be verified using the calculated interface costs rather than an alternate physical path.

6.4 Network Scalability

As the number of routers and departments increases, RIP's 15-hop limit and full-table broadcasts would quickly become a bottleneck for a growing university network with many buildings. OSPF's link-state, area-based design and lack of a hop-count ceiling make it noticeably better suited to a campus network that is expected to expand over time.

Conclusion

This project successfully designed, implemented, and tested a multi-department university network connecting the Academic, Administrative, and Research buildings using two different dynamic routing protocols, RIP and OSPF, in Cisco Packet Tracer.

The comparative study clearly showed that while RIP is simple to configure and adequate for very small, flat networks, it suffers from slow convergence, a restrictive 15-hop limit, and inefficient periodic full-table broadcasts. OSPF, on the other hand, demonstrated faster convergence through event-triggered link-state updates, a more accurate cost-based path metric, and no hard limit on network diameter, making it far better suited to a growing, multi-building university environment.

Based on the routing table formation, convergence behaviour, hop-count constraints, and scalability observed during simulation, OSPF is recommended as the preferred dynamic routing protocol for the university's multi-department network, with RIP being suitable only for very small branch segments or as a fallback/legacy protocol.

Future Scope

The current design can be extended in several directions to more closely reflect a real, production-grade university network:

- Implement multi-area OSPF (Area 0 backbone with departmental areas, e.g., Area 1 for Academic, Area 2 for Admin, Area 3 for Research) to further improve scalability and limit LSA flooding domains.
- Apply Variable Length Subnet Masking (VLSM) to the backbone links (e.g., /30 subnets) instead of /24, to conserve IP address space.
- Introduce redundant links and HSRP/VRRP for default-gateway and link failover at each department LAN.
- Add access-control lists (ACLs) and inter-VLAN routing to segment traffic between student, faculty, and administrative devices within each department.
- Extend the design with IPv6 dual-stack addressing (OSPFv3) to prepare the university network for future IPv6 migration.
- Integrate EIGRP or BGP for comparison if the university network were to connect to an external ISP or a second campus, and evaluate hybrid/exterior routing scenarios.
- Deploy network monitoring (SNMP/NetFlow) and simulate real traffic loads to study protocol performance under congestion, not just topology-change convergence.

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